



Anand Narasimhan
Computer Science & Engineering
Indian Institute of Technology Bombay

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B.Tech.
Gender: Male
DOB: 17/06/2003

Examination	University	Institute	Year	CPI / %
Graduation	IIT Bombay	IIT Bombay	2025	
Intermediate	CBSE	MVM Sr. Sec. School, Chetpet, Chennai	2021	98.20%
Matriculation	CBSE	MVM Sr. Sec. School, Chetpet, Chennai	2019	96.00%

Pursuing a **Honours** degree from the **CSE Department**, IIT Bombay

SCHOLASTIC ACHIEVEMENTS

- Secured **All India Rank 17** and was the **Tamil Nadu State Topper** in **JEE-Advanced** (2021)
- Secured **Rank 28** and **Rank 32** in the Indian National Olympiads in Physics and Chemistry respectively and attended the *Orientation cum Selection Camps* for both subjects (2021)
- Awarded the **KVPY Fellowship in Basic Sciences** twice by the Indian Government, by securing **National Rank 56** in the SX stream ('20) and **National Rank 99** in the SA stream ('19). (2019-20)
- Received an **AP Scholar with Distinction** award for scoring **5/5** in **7** different **AP** (Advanced Placement) exams across **Computer Science, Math, Physics, Chemistry and English Composition** (2017-2020)
- Achieved a **SAT** score of **1550/1600** with a perfect score of **800/800** in the **Math** section (2019)
- Selected as an **NMS**(National Merit Scholar) Finalist from over **1.6 million** candidates by virtue of scoring **1490/1520** on the **PSAT/NMSQT** (Preliminary Scholastic Aptitude Test/NMS Qualifying Test) (2019)

RESEARCH EXPERIENCE

Approximating DFAs with RNNs | Summer Research Internship

(Summer '23)

Guide : Prof. Eric Fabre, INRIA, Rennes

- Investigated the problem of approximating **DFAs** using **RNNs** trained on the corresponding regular language
- Found significant clustering among the **RNN's hidden states**, similar to the discrete states of an automaton
- Experimented with **Bi-RNNs**, **LSTMs** to improve clustering and **t-SNE** to plot higher dimensional states in 2D

Graph Hashing | R&D Project

(January '24 - July '24)

Prof. Abir De, CSE, IIT Bombay

- Worked on designing a **Locality Sensitive Hashing** (LSH) procedure on a novel similarity score between a query and a corpus document, represented as **graph embeddings**, to perform **subgraph matching** in the document
- Analysed existing LSH mechanisms and **Fourier-transform** based hashing methods for set containment search
- Incorporated** our **LSH mechanism** in the code for **ISONET**, a subgraph-matching based retrieval model

Machine Learning Intern | Winter Internship

(Winter '22)

ZLabs, Zoho Corporation

- Explored various **ML** and **DL** based models and benchmarks in the field of **Document Intelligence**
- Implemented the **RewriteNet** model used in **Scene Text Editing** to produce **realistic synthetic data**
- Studied the problem of fitting **log-linear** models to data where the dependent variable can be **zero** or **negative** and implemented several estimators such as **iterative OLS** (iOLS), **i2SLS**, **PPML** and **Multiplicative Poisson**

KEY PROJECTS

Specformer - Spectral GNNs meet Transformers | Course Project

(Autumn '23)

Instructor: Prof Abir De - Learning with Graphs

- Implemented the **Specformer** paper, which proposes a **Transformer**-based architecture to improve **Spectral GNNs**, applying self-attention on the set of eigenvalues of the Laplacian to learn better **Spectral Filters**
- Worked on implementing **Deformed Attention** to **sparsify** the attention matrix, reducing the model complexity

Fast Chat | Course Project

(Autumn '22)

Instructor: Prof. Kavi Arya - Software Systems Lab

- Constructed an interactive messaging network amongst numerous clients, with servers regulating the process
- Implemented functionalities such as **group chats**, **sending images** and an **admin** role, with **E2E encryption** between users. Experimented with different load balancing strategies for multiple servers as well
- Maintained a database using **PostgreSQL** keeping track of user data, public keys and undelivered messages

Diffusion Models in Anomaly Detection | Course Project

(Spring '24)

Instructor : Prof Sunita Sarawagi - Advanced Machine Learning

- Conducted analysis of advanced state of the art **Diffusion Models** applied in the field of image anomaly detection
- Implemented and analysed variations of the **DiffusionAD** and **D3AD** models, modifying the **norm-guided** single **denoising step**, along with using a **KNN-based** method for choosing the number of noising steps on the image

SCLP: A Language Processor for a Small C-like Language | *Course Project* (Spring '24)

Instructor: Prof. Uday Khedker - Compilers Lab

- Built a compiler that supported features in **C** including **conditionals**, **control flow and recursion**
- Used tools like **lex** and **yacc** for scanning and bottom-up parsing and generated intermediate representations like ASTs, Three Address Codes, Register Transfer Language and MIPS assembly code with register allocation

Knowledge Graph-based Question Answering System | *Course Project* (Autumn '23)

Instructor : Prof Soumen Chakrabarti, CSE, IIT Bombay

- Developed a **Knowledge Graph-based QA System** leveraging a dataset containing the top 1000 **IMDB** movies
- **Cleaned** the data, created **graph embeddings** and picked the **GPT-3** model to handle raw questions.
- Used **LangChain** to integrate the graph embeddings and the model into a **question answering interface**

Railway Planner | *Course Project* (Autumn '22)

Instructor: Prof. Supratik Chakraborty - Data Structures and Algorithms Lab

- Structured a railway planner in C++ using data structures such as BSTs, Dictionaries, Heaps, Tries and AVLs
- Built an auto-completion feature to aid the search for station names by using the **Trie** data structure
- Adapted quicksort on linked list to list departures sorted by time and priority queue to fetch list of top-rated trains

Manifold Representation Learning | *Course Project* (Autumn '23)

Instructor: Prof. Preethi Jyothi - Artificial Intelligence and Machine Learning

- Exploring various approaches to non-linear dimensionality reduction using **Manifold Learning**, like **t-SNE**, **LLE**
- Analyzing and implementing a particular approach, **Markov-Lipschitz** Deep Learning, using **Markov Random Fields** and local **bi-Lipschitz** continuity to obtain better **local geometry preservation** and **robustness**

Automatic Speech Recognition | *Course Project* (Sept 2023-Nov 2023)

Instructor: Prof. Preethi Jyothi - Automatic Speech Recognition

- Utilized **Whisper** model for speech recognition task on Code-Switched Hinglish dataset, achieving **WER** of 0.493
- **Presented LatticeST**, a paper on incorporating Lattice inputs to purely **self-attentional models**
- Simulated a modular framework for global normalization using **WFSAs** for **Streaming Speech Recognition** task

Quantum Computing, ML and Quantum ML | *Seasons of Code* (May 2022 - July 2022)

Web and Coding Club, IIT Bombay

- Implemented **Quantum Computing Algorithms** such as Quantum Teleportation, Deutsch-Jozsa, Quantum Fourier Transforms, Phase Estimation, Shor's Algorithm, Grover Search using the **Qiskit** SDK
- Modified a **Quantum General Adversarial Network**, coded in the **PennyLane** Python library with **PyTorch** support, that generates hand written images of zero, using the recently proposed **Patch Method** for the network

POSITIONS OF RESPONSIBILITIES

Manager | *Maths and Physics Club, IIT Bombay* (May 2023 - May 2024)

- Led a team of **8 people** in organising Math and Physics events for enthusiasts of the institute, such as quizzes, competitions, group discussions, trips and lab visits, while managing day-to-day logistics and social media pages

Institute Student Mentor | *Student Mentorship Program, IIT Bombay* (July '24 - Present)

- Mentoring 12 undergraduate freshmen through all academic and personal issues while transitioning to college life
- Selected in a **team of 150 mentors** from a pool of 400+ applicants after extensive interviews and peer reviews

Teaching Assistant | *CSE and Math Departments, IIT Bombay* (March 2023-Present)

- **CSE:** Handling labs, designing exams, creating additional course material and solving doubts of 200+ students in two courses, **Abstractions and Paradigms of Programming**, and **Introduction to Machine Learning**.
- **Mathematics:** Conducting weekly problem-solving sessions in Linear Algebra and Calculus for a class of 50.

Class Representative | *CSE Batch of 2025 - Second Year* (July 2022 - May 2023)

- Acted as the primary point of communication between the professors and a batch of 194, for all course logistics

RELEVANT COURSES UNDERTAKEN AT IITB

Computer Science

Advanced Machine Learning, Automatic Speech Recognition, Learning with Graphs, Modern Information Retrieval, Algebraic Automata Theory, Computational Complexity, Formal Methods in ML, Extremal Combinatorics, Quantum Computing
Basic Algebra, Real Analysis, Calculus, Linear Algebra, Differential Equations

Mathematics

TECHNICAL SKILLS

Languages/Software

C/C++, Python, Lisp (Scheme), Haskell, Bash, Java, Prolog, Git, $\text{\LaTeX}$

Data Science

PyTorch, NumPy, Pandas, Matplotlib, Seaborn, TensorFlow, Keras, Qiskit, PennyLane

Web Dev & Databases

HTML, CSS, JavaScript, Bootstrap, SQL, Flask

EXTRACURRICULAR ACTIVITIES

- **Winner of Bazinga Math Quiz 2022** organised by Maths and Physics Club, IIT Bombay (2022)
- Participated in the **IIT Bombay Scrabble Open**, a **SAI rated** event of top-rated players. (2022-2023)
- Trained in *Carnatic Music, Vocals and Keyboard* and have given multiple performances (2010 - Present)